

Calibration-Free fp8 Quantization for Transformers: A Shannon-Prime Experiment Design

A. Knack (Shannon-Prime Project) Companion systems paper. Draft v0.1 — 2026-05-16

Abstract

This companion paper specifies the experiment that probes the single highest-leverage prediction of the Shannon-Prime framework: that fp8 quantization on a CM-encoded transformer state requires no calibration. The prediction is a direct consequence of the Frobenius Quantization Theorem (Theorem 2 of the theory companion [SP-Theory-Companion 2026]); this paper presents the experimental design that tests it on a Phi-3 reference model, with the full implementation hooks already present in the Shannon-Prime engine. We describe four configurations (baseline fp16, SP-fp8 calibration-free, GPTQ-fp8 calibrated, AWQ-fp8 calibrated) and the evaluation harness, perplexity targets, and pass/fail criteria. The expected outcome is that SP-fp8 matches the calibrated baselines without any calibration data. We include implementation notes, a reproducible run script outline, and a discussion of how the result, if positive, scales to fp4.

1. The Question

Standard transformer quantization treats the model’s weight and activation tensors as floating-point arrays and rounds them to a lower-precision representation. The rounding introduces errors in the multiplicative relations the model has learned during training, and post-training calibration is required to fit a per-tensor scale factor that approximately restores those relations. GPTQ, AWQ, SmoothQuant, and QServe all follow this pattern; each requires a calibration pass on representative data (typically ~1000 samples) to recover acceptable accuracy.

Shannon-Prime predicts something different. Its theoretical content (Theorem 2 of [SP-Theory-Companion 2026]) is that on a CM elliptic curve over $\mathbb{Q}(\sqrt{-163})$, the Frobenius endomorphism φ_p commutes with the layer endomorphisms. If the model’s state is encoded on this curve — as SP encodes the KV cache and (in this experiment) the weights — then reduction to lower precision is Frobenius application rather than rounding, and the multiplicative relations are preserved exactly.

Question. *Does SP-fp8 on a Phi-3 model recover vanilla-fp16 perplexity to within noise, with no calibration data?*

A positive answer would deliver one of the strongest practical results in the field: a quantization scheme requiring no calibration pass, hence shippable to edge devices without device-specific data.

2. Setup

2.1 Reference Model

Phi-3 was chosen for three reasons:

1. **It is the SP model-pack’s calibrated reference.** Phi-3 currently passes the SP calibration ledger at $\Delta\text{PPL} = +2.44\%$ vs vanilla under SP-base compression. Starting from a known-calibrated SP target removes one source of variance.
2. **It is small enough to bench quickly.** 3.8B parameters at fp16 fits on a single A100; the bench runs in under an hour per configuration.
3. **It does not exhibit the Qwen3 edge-fail artifact.** The Qwen3 architecture’s gated-attention + mRoPE-mode-8 combination breaks the SP encoding in a known way; Phi-3 does not. Using Phi-3 eliminates that confound.

2.2 Bench Corpora

Two corpora at three context lengths each:

Corpus	Description	Tokens
WikiText-103	Standard perplexity bench	245M
The Stack v2 (filtered)	Code completion bench	100M

Context lengths: 512, 2048, 8192. The 8192 condition exercises the long-context KV cache path where SP compression is most impactful.

2.3 Configurations

Four configurations, all evaluated on the same corpus + context combinations:

Config	Quant	Calibration	Notes
A. Baseline	fp16	—	Ground truth
B. SP-fp8 calibration-free	SP-fp8 ($p = 11, \varphi_p^8$)	None	The hypothesis
C. GPTQ-fp8	fp8	~1000 samples	Industry baseline
D. AWQ-fp8	fp8	~1000 samples	Industry baseline

The configuration that is the focus is B. The hypothesis is that B matches A within noise ($|\Delta\text{PPL}| < 0.5\%$) and matches C/D within noise ($|\Delta\text{PPL}_{B,C}| < 0.3\%$).

3. Why fp8, Why $p = 11$, Why φ_p^8

The Frobenius Quantization Theorem (Theorem 2 of [SP-Theory-Companion 2026]) implements $\text{fp16} \rightarrow \text{fp}q$ as iterated Frobenius φ_p^{16-q} . For $q = 8$, this is φ_p^8 . The prime p is constrained by $p^q \leq 2^{16}$, i.e., $p^8 \leq 2^{16}$, giving $p \leq 2^2 = 4$, which is too small to be useful at $q = 8$.

The resolution is to relax the constraint: at $q = 8$ we have an entire byte of representation, and p can be chosen up to $2^8 = 256$. Setting $p = 11$ gives a comfortable margin while keeping the trace of Frobenius a_{11} in a numerically friendly range. For E the CM curve over $\mathbb{Q}(\sqrt{-163})$ reduced mod 11, a_{11} can be computed from the Hilbert class polynomial; the explicit value is $a_{11} = -6$ (good ordinary reduction). The Frobenius polynomial is then $\varphi_{11}^2 + 6\varphi_{11} + 11 = 0$ in $\text{End}(E_{11})$.

The number of points: $\#E_{11}(\mathbb{F}_{11}) = 11 + 1 - a_{11} = 18$, well within the Hasse–Weil bound of $11 + 1 + 2\sqrt{11} \approx 18.63$.

At $q = 8$, per-coordinate information capacity is $\log_2 18 \approx 4.17$ bits. This is lower than the nominal “8 bits” of fp8 because the CM encoding uses the upper bits for spinor and Möbius coefficients. Empirically this matches the effective bit-width of well-quantized fp8 LLM weights as reported in [Liu et al. 2024 KIVI].

4. Implementation

4.1 Existing Hooks

The Shannon-Prime engine already contains:

- `--hier-ternary-mask` and `--hier-res-bits-v` flags for ternary skeleton compression.
- VHT2 + Möbius + spinor + square-free chain across all four backends.
- The fp8 quantization implementation in the engine path (used in production but currently calibrated).
- Per-architecture compression-default registry (model-pack scaffold) — phi3 row already exists and is calibrated.

What is needed to run the experiment:

1. **A `--frobenius-quant` flag** that toggles the Frobenius reduction in place of the calibrated fp8 path.
2. **A unit test** verifying that φ_{11}^8 produces output bit-identical to direct reduction mod 11^8 on the CM-encoded state.
3. **A bench harness modification** to disable the calibration-data path for config B.

Estimated implementation: 1–2 days at the engine level. The math is fully specified by Theorem 2.

4.2 Outline of the Run Script

```
# Pseudocode for the bench harness
for cfg in [A_baseline_fp16, B_sp_fp8_calfree, C_gptq_fp8, D_awq_fp8]:
    model = load_phi3(cfg.quantization)
    for corpus in [wikitext103, stackv2]:
        for ctx in [512, 2048, 8192]:
            ppl = eval_perplexity(model, corpus, ctx)
            tokens_per_sec = bench_throughput(model, corpus, ctx)
            log(cfg, corpus, ctx, ppl, tokens_per_sec)
```

All four configs use the same bench loader and the same tokenization; only the quantization differs.

4.3 Pass/Fail Criteria

Comparison	Criterion
B vs A (calibration-free fp8 vs baseline fp16)	$ \Delta\text{PPL} < 0.5\%$ on WikiText-103 at all contexts
B vs C (SP-fp8 vs GPTQ-fp8)	$ \Delta\text{PPL}_{B,C} < 0.3\%$ at all contexts
B vs D (SP-fp8 vs AWQ-fp8)	$ \Delta\text{PPL}_{B,D} < 0.3\%$ at all contexts
B throughput	within 5% of A throughput; SP-fp8 is not expected to be faster than calibrated fp8
Robustness	B’s relative PPL gap does not increase with context length

If all four criteria pass, the framework’s central practical prediction is validated and SP-fp8 is deployable as a calibration-free quantization.

5. Expected Outcome and Risk Analysis

5.1 Expected Outcome

Theorem 2 predicts that B matches A within numerical noise. The framework attributes any residual gap to:

- The Frobenius trace a_p being nonzero (predicted systematic drift of $\approx a_p/p \approx 0.5\%$ at $p = 11$); the experiment can quantify this.
- Bench-scaffold artifacts (the fp32 reduction step in the engine bench loop introduces small additional error; see the A100 result in [SP-Systems 2026]).
- Tokenizer interaction with the CRT-sharded LM head (not exercised by config A but exercised by SP encoding).

The expected magnitude of $|\Delta\text{PPL}_{B,A}|$ is in the range 0.1% to 0.5%. Anything below 1% is a positive result; anything below 0.3% is a strong positive.

5.2 What a Negative Result Would Indicate

A failure of B to match A would localize one of the following:

1. The CM realization of Phi-3’s layer endomorphisms is imperfect — the Endomorphism Realization (Theorem 1 of [SP-Theory 2026]) holds in principle but does not yet match the empirical model in implementation.
 2. The Frobenius reduction is implemented incorrectly in the engine.
 3. The framework’s choice of curve E over $\mathbb{Q}(\sqrt{-163})$ does not match the actual hidden-state geometry of trained Phi-3.
- (1) is debuggable by mechanistic interpretability of layer endomorphisms; (2) is a code review; (3) would require revisiting the framework’s curve choice. Of these, (3) is the lowest-probability and the most consequential.

5.3 What a Positive Result Enables

A positive result delivers:

1. **Calibration-free fp8 deployment.** Drop-in fp8 for any Phi-3-class model with no calibration corpus required.
2. **A bridge to fp4.** Theorem 2 with $q = 4$ requires $p \leq 11$, and the SP-fp4 numerical information ceiling of ≈ 4.07 bits/coord (Corollary 2.2 of the theory companion) matches the empirical fp4 capacity. The same experiment, repeated at $q = 4$ with appropriate p , would be the natural next step.
3. **A target for ARM/Hexagon production.** Phone-class targets currently quantize via runtime-calibrated routines. Calibration-free quantization removes a major deployment friction.

6. Implementation Timeline

Phase	Deliverable	Estimated effort
1. Engine hooks	<code>--frobenius-quant</code> flag, unit test for φ_{11}^8	1–2 days
2. Bench harness	Modified eval loop for the four configs	1 day
3. Reference runs	All four configs at all six (corpus $\times$ context) combinations	1 day on A100
4. Analysis + writeup	Result table, plot of PPL vs context, regression to the framework’s predictions	2 days
5. Companion v0.2	Incorporate results into this paper as §5 (Results)	1 day

Total: $\sim$ 1 week of engine + bench work, assuming an A100 is available for the reference runs.

7. Connection to the Broader SP Framework

This experiment is one component of a larger program. The Shannon-Prime framework predicts five additional implementations, each with its own experimental signature:

- **Stern–Brocot RoPE** ([SP-Theory 2026, §9.1]): replace RoPE base 10,000 with φ ; bench on long-context tasks. Smallest implementation cost of any extension.
- **Weil pairing attention** ([SP-Theory 2026, §9.2]): replace softmax-of-dot-product with the Weil pairing e_n . Highest research upside.
- **Hecke-eigenform embeddings** ([SP-Theory 2026, §9.3]): replace learned embedding with a Hecke basis. Longest path to a number.
- **L-function activation oracle** ([SP-Theory 2026, §9.4]): predict FFN firing as L-function coefficients.
- **LLL KV write** ([SP-Theory 2026, §9.5]): make the KV archive an LLL-reduced lattice basis.

Each of these is the subject of a future SP companion paper. The present experiment — calibration-free fp8 — is first in line because it is the smallest implementation step that probes the strongest theoretical prediction.

8. Conclusion

Shannon-Prime’s Frobenius Quantization Theorem predicts that fp8 quantization on a CM-encoded transformer state requires no calibration data. The present companion paper has specified the experimental design that tests this prediction on a Phi-3 reference model. The result, if positive, validates the framework’s strongest practical claim and opens a path to fp4 and edge-device deployment without calibration. The experiment is small (1 week of engineering + benchmarking), the implementation hooks are already in place across the SP engine and llama.cpp fork, and the framework’s predicted outcome is sharp and falsifiable.

References

- Companion theory: “Two Theorems for Shannon-Prime Compression: Hasse–Weil as the Shannon Limit, and Frobenius as Quantization.” 2026.
- Full theory: “Shannon-Prime: A CM-Elliptic-Curve Framework for Transformer Computation.” 2026.
- Full systems: “Shannon-Prime: Provably Exact KV-Cache Compression and Per-Layer Acceleration on Standard Inference Stacks.” 2026.
- Liu, Z. *et al.* “KIVI: A Tuning-Free Asymmetric 2bit Quantization for KV Cache.” 2024.
- Frantar, E. *et al.* “GPTQ: Accurate Post-Training Quantization for Generative Pre-trained Transformers.” 2023.
- Lin, J. *et al.* “AWQ: Activation-aware Weight Quantization for LLM Compression and Acceleration.” 2024.
- Phi-3 Technical Report, Microsoft, 2024.